Cross-Corpus Evaluation of Multiple Pose Estimation Methods for Sign Language: Towards Metrics for Accurate Sign Representation

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1. Introduction

Sign languages (SLs) are natural languages used by Deaf communities worldwide. Each sign language has its own lexicon and grammatical structure. They are expressed through manual and nonmanual features, including hand and body movements, and facial expressions. Together, these features convey lexical and grammatical information, and slight variations in their form can alter the message [1].

Automatically processing sign language is challenging due to its complex spatial and temporal structure. Automatic Sign Language Processing (A-SLP) addresses this challenge through three core tasks: Sign Language Recognition (SLR), Sign Language Translation (SLT), and Sign Language Production (SLP). SLR maps visual signs to symbolic representations such as glosses; SLT translates sign language into written text; and SLP performs the reverse direction, generating sign language from written text [2].

Pose estimation is widely used across A-SLP to represent the signer's body configuration and motion, although it is not required by every recognition, translation, or production system. In recognition and translation, pose information can be extracted from video as skeletal keypoints, while production systems can use pose sequences as an intermediate representation for generating signs. Compared with raw video, these representations focus more directly on body structure and motion. A common approach for extracting them is Markerless Pose Estimation (MPE), which estimates human pose directly from images or video without physical markers. As illustrated in Figure 1, the methods considered in this work are divided according to their output representation into Keypoint Estimation, which represents the body through anatomical landmarks, and Human Mesh Recovery (HMR), which reconstructs a 3D representation of the human body [3].

Keypoint Estimation represents human pose using sparse 2D or 3D joint coordinates, providing a compact representation of body configuration. In contrast, HMR reconstructs a dense 3D surface representation of the body. Parametric models such as SMPL and its expressive extension, SMPL-X, encode body shape and pose, with SMPL-X additionally modeling articulated hands and facial expressions. This richer representation captures geometric details that are not available from joint coordinates alone, particularly fine-grained hand articulation relevant to sign language applications [4].

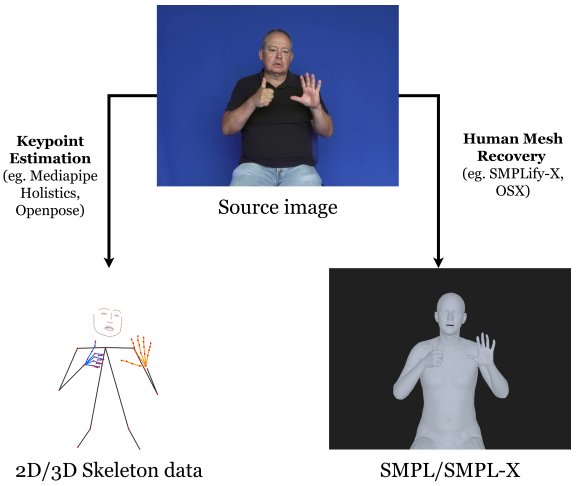


Figure 1: Two primary approaches to Markerless Pose Estimation: Keypoint Estimation, which outputs computationally lightweight 2D/3D skeleton data, and Human Mesh Recovery, which produces full volumetric 3D meshes such as SMPL or SMPL-X to preserve structural data.

In HMR, reconstruction quality is commonly evaluated using the Mean Per Joint Position Error (MPJPE) and Per-Vertex Error (PVE). MPJPE measures the average Euclidean distance between predicted and ground-truth 3D joints after root alignment, while PVE measures the average distance between corresponding predicted and ground-truth mesh vertices [5]. Both metrics therefore require 3D ground-truth annotations, limiting their application to sign language datasets without reference joints or meshes. As such annotations remain limited in A-SLP datasets, HMR predictions often cannot be directly evaluated using standard joint- or vertex-based errors, making objective assessment of reconstruction accuracy challenging [2].

To address this limitation, this work proposes an evaluation methodology for markerless body, hand, and face reconstruction in sign language. The approach combines 2D automatic references with specialized metrics, including fingertip positional error (Tip-NME) and digit extension matching (HCA-2D). By evaluating position, extension, and coverage separately, the framework en-

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ables HMR models to be assessed directly on in-the-wild sign language corpora without requiring 3D ground-truth annotations.

2. Preliminaries

When 3D ground-truth annotations are unavailable, pose estimates can instead be evaluated in the image plane using 2D keypoint metrics. Object Keypoint Similarity (OKS) is a widely used metric for evaluating 2D keypoint localization. It measures the similarity between predicted and ground-truth keypoints while accounting for person scale, keypoint type, and visibility [6]. It is defined as:

$$OKS = \frac{\sum_i \exp\left(-\frac{d_i^2}{2s^2\kappa_i^2}\right)\delta(v_i > 0)}{\sum_i \delta(v_i > 0)} \tag{1}$$

where d_i is the Euclidean distance between the predicted and ground-truth location of keypoint i, s represents the scale of the person, κ_i is a keypoint-specific constant controlling the distance falloff, and $\delta(v_i > 0)$ restricts the calculation to annotated keypoints. OKS ranges from 0 to 1, with higher values indicating greater spatial agreement.

As a complementary measure, Normalized Mean Error (NME) quantifies the average keypoint localization error relative to a reference scale, reducing the effect of differences in subject size or image resolution [6]. It is defined as:

$$NME = \frac{1}{N}\sum_{i=1}^{N}\frac{d_i}{d_{\text{ref}}} \tag{2}$$

where N is the number of evaluated keypoints, d_i is the Euclidean distance between predicted and ground-truth keypoints, and d_{ref} is the chosen normalization distance, such as a bounding-box dimension or an anatomical reference distance. Lower NME values indicate smaller normalized localization errors.

These metrics can also be computed separately for each hand. Following part-specific evaluation protocols used in whole-body HMR, which report errors independently for the body, left hand, and right hand [5], we calculate *Left-Hand OKS*, *Left-Hand NME*, *Right-Hand OKS*, and *Right-Hand NME*. Given the linguistic importance of hand configuration and movement in sign language, these hand-specific metrics provide a more targeted assessment for A-SLP applications [7].

However, OKS and NME still aggregate localization accuracy across multiple keypoints, so errors in individual fingers can be diluted by accurate localization elsewhere in the hand. Moreover, when automatically estimated 2D keypoints are used as references, errors in those estimates propagate into the resulting scores. Figure 2 illustrates this limitation: PyMAF-X reproduces the source's extended index and middle fingers but obtains the lowest OKS (0.1813) and highest NME (0.0994). PEAR omits the middle finger yet achieves a lower NME (0.0854), while Multi-HMR extends an additional finger but receives the highest OKS (0.3004). Thus, higher aggregate spatial accuracy does not necessarily correspond to correct finger configuration, which can distinguish meaning in sign language.

3. Methods

3.1. Search and selection of HMR methods

A structured search combined the terms *human mesh recovery*, *whole-body reconstruction*, *SMPL-X*, and *hand reconstruction* in Google Scholar, IEEE Xplore, CVF Open Access, ACM Digital Library, SpringerLink, and arXiv. Reference screening of recent HMR surveys complemented the search [5, 8]. The scope covered reconstruction from monocular RGB images and video, including integrated systems that combine specialized estimators. Preprints and subsequent publications describing the same method were treated as one candidate.

Inclusion criteria. Titles, abstracts, and bibliographic metadata were screened for all of the following:

- Publication or initial public release from January 2023 onward.
- Estimation of human pose and surface geometry from RGB observations.
- Whole-body reconstruction with articulated hands.

Exclusion criteria. Complete articles, supplementary material, and official model documentation were assessed to exclude methods that:

- Required synchronized views, measured multicamera calibration, depth, markers, or reference 3D poses at inference.
- Lacked a parametric representation of the body, articulated hands, and facial expression.
- Had no pretrained model for the complete reconstruction procedure available for direct research use.

Monocular video methods were eligible because temporal context is available in sign language recordings. Eligible representations included SMPL-X and expressive extensions combining its body and hands with a separate facial model.

For each retained method, we recorded the input type, reconstruction strategy, output representation, and published evaluation protocol. Reported errors were associated with their dataset, model variant, anatomical region, alignment, and units. Missing benchmark values were left unreported, and component-level scores were not attributed to an integrated system.

3.2. HMR video reconstruction from sign language corpora

The reconstruction protocol uses RGB sequences from PHOENIX-2014-T, CSL-Daily [9, 10], and a Chilean Sign Language (LSCh) collection, OF-HMR. Each reconstruction receives a single camera view. Image sequences were assembled into videos with one source image per video frame, preserving chronological order and spatial resolution. PHOENIX-2014-T uses 210 × 260-pixel frames

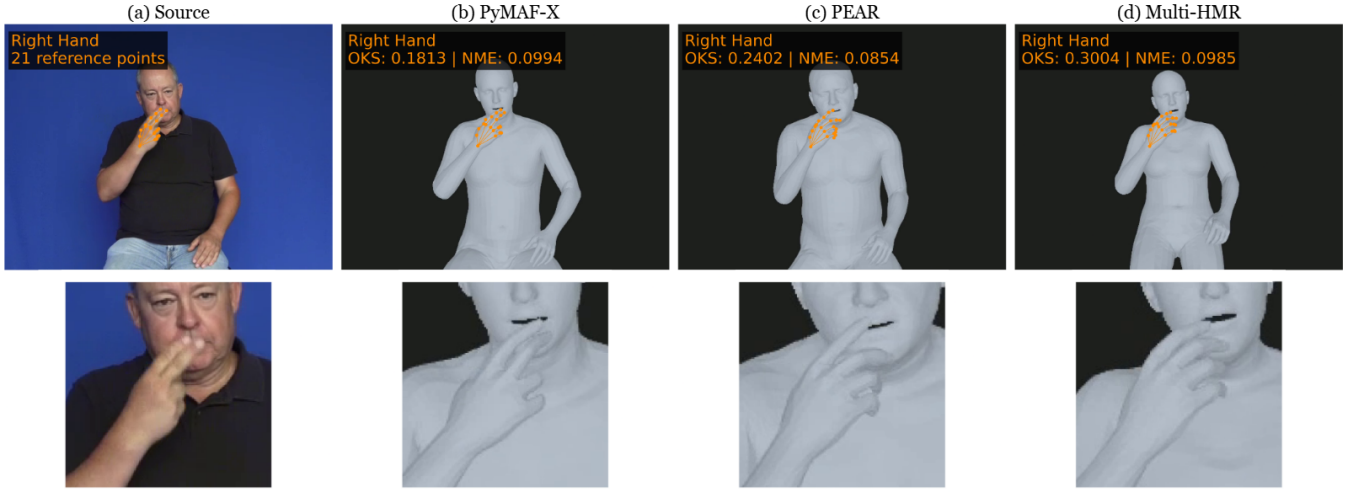


Figure 2: Qualitative comparison of right-hand reconstruction across HMR models. (a) Source image with 21 right-hand reference keypoints. (b–d) Reconstructions produced by PyMAF-X, PEAR, and Multi-HMR, respectively, with their corresponding OKS and NME scores. The enlarged hand regions highlight differences in finger configuration that are not fully reflected by the aggregate spatial metrics.

at 25 frames per second; CSL-Daily uses 512 × 512-pixel frames at 30 frames per second. The OF-HMR sequences use 1440 × 1080-pixel frames at 30 frames per second.

Each input video was processed by the selected HMR methods using pretrained models without further training on the corpus. Image-based methods reconstructed each frame independently, while video-based methods used their temporal context. The resulting meshes were rendered using each method's estimated camera to reproduce the source viewpoint and signer scale. All methods used the same neutral material, lighting, and uniform background, with the source frame count, resolution, and frame rate preserved.

Mean reprojection error (MRE) compares RTMW-x keypoints detected in each original image with anatomically corresponding mesh keypoints projected into that image using the camera and placement of the rendered reconstruction [11]. RTMW-x provides automatic whole-body references; its use for body and finger localization is also discussed by Diallo and Akhloufi [12]. Source observations must have confidence at least 0.5 and lie within the image. Hand observations with implausible bone lengths, spatial extent, or wrist correspondence relative to shoulder width are excluded. The same reference observations are used for all ten methods.

Camera and placement correction addressed apparent viewpoint or distance changes inconsistent with the stationary recording setup. Corrections jointly estimated constant camera intrinsics and a source-driven translation trajectory, preserving articulated pose, orientation, and body dimensions. Selection considered regional error, agreement with source motion, and full-head visibility.

Evaluation uses 99 landmarks available in every method: six main-body landmarks (the left and right shoulders, elbows, and wrists), 51 internal facial landmarks, and 21 landmarks per hand. Body MRE excludes all facial landmarks and articulated finger landmarks. Facial contour landmarks and the five coarse head landmarks are excluded from every reported measure, so overall MRE covers exactly the union of the four anatomical regions. At frame t, for region r, the error is

$$\mathrm{MRE}_{r,t} = \frac{1}{|\mathcal{V}_{r,t}|} \sum_{i \in \mathcal{V}_{r,t}} \|\hat{\mathbf{x}}_{i,t} - \mathbf{x}_{i,t}\|_2, \tag{3}$$

where $\mathcal{V}_{r,t}$ contains eligible corresponding landmarks shared by all ten methods, $\mathbf{x}_{i,t}$ is the source reference, and $\hat{\mathbf{x}}_{i,t}$ is the projected reconstruction. Errors are measured in source-image pixels. Frame errors are averaged within each sequence, followed by an equal-sequence mean and sample standard deviation. Frames without an eligible observation in a region do not contribute to that regional mean. Only sequences with complete reconstructions from all ten methods enter the comparison. Correction selection and evaluation use the same automatic references, so the values describe within-sequence 2D agreement rather than independent 3D accuracy or linguistic fidelity.

4. Results

4.1. Search and selection of HMR methods

The selection retained ten monocular reconstruction methods: PyMAF-X, OSX, SMPLer-X, HybrIK-X, SMPLest-X, PEAR, AiOS, Multi-HMR, WHAC, and Video-to-SMPL-X. The first eight estimate expressive meshes from individual images, whereas WHAC and Video-to-SMPL-X operate on monocular video. Table 1 summarizes published mesh errors available for a majority of the selected methods under the same benchmark and metric.

SMPLify-X was excluded at metadata screening because its 2019 publication predates the inclusion period [13]. Assessment of the complete reconstruction procedures excluded MAMMA and the reviewed multiview configuration of MKA because they require calibrated observations from multiple cameras [14, 15]. Mamba-HMR predicts non-parametric meshes, while SAM 3D Body does not estimate facial expression, so neither met the expressive parametric output requirement [16, 17]. RoboSMPLX did not provide the required pretrained whole-body model, and SMFLIX offered its model only

upon request; both therefore failed the direct-availability requirement [18, 19].

Table 1: Published errors of the ten selected HMR methods (mm; lower is better). AGORA-test reports overall MVE without the detection penalty used by NMVE. EHF reports full-mesh PVE and hand PA-PVE, with PA denoting Procrustes alignment. Dashes indicate that the corresponding result is not reported in the cited source.

	AGORA-test	EHF	
Method	MVE	PVE	Hand PA-PVE
PyMAF-X [20]	125.7	64.9	10.2
OSX [21]	122.8	70.8	15.9
SMPLer-X [22]	99.7	65.4	15.0
HybrIK-X [23]	112.1	—	—
SMPLest-X [24]	86.6	49.7	11.4
PEAR [25]	—	—	12.8
AiOS [26]	91.9	45.4	12.8
Multi-HMR [27]	95.9	42.0	10.8
WHAC [28]	—	—	—
Video-to-SMPL-X [29]	—	—	—

PyMAF-X: HR48 for EHF. SMPLer-X: L20; SMPLest-X: H40; both use AGORA fine-tuning for AGORA-test. Multi-HMR: ViT-L, 896×896, without camera intrinsics in the reported evaluation. AiOS: detection threshold 0.5 for AGORA. PEAR: ViT-B, 256×192. Model variants and training data differ across studies.

The image-based results show that whole-body and hand accuracy need not favor the same method. On EHF, Multi-HMR reports a lower full-mesh PVE than PyMAF-X (42.0 versus 64.9 mm), whereas PyMAF-X reports a lower hand PA-PVE (10.2 versus 10.8 mm). These measures evaluate different regions and remove different geometric discrepancies. Neither directly measures whether the reconstructed fingers preserve a sign's hand configuration.

The video methods address complementary aspects of reconstruction. WHAC jointly estimates human motion and camera trajectories, reporting 142.2 mm WA-MPJPE over world-aligned 100-frame segments on EMDB2 [28]. Video-to-SMPL-X combines specialized body, hand, and facial estimators [29]. It has no reported complete-system evaluation on AGORA or EHF; its component-level scores cannot establish joint reconstruction accuracy.

4.2. HMR video reconstruction from sign language corpora

The comparison includes 42 PHOENIX-2014-T development sequences (4,568 frames), 29 CSL-Daily training sequences (1,596 frames), and eight LSCh sequences from OF-HMR (838 frames). Each sequence has a complete reconstruction from all ten methods, yielding 790 reconstructions over 7,002 source frames. Tables 2, 3, and 4 report overall, body, face, left-hand, and right-hand MRE separately for each dataset. The left-hand results for CSL-Daily use the 26 sequences with eligible observations; the other three have no left-hand measurement and do not contribute to that regional mean.

On PHOENIX-2014-T, PEAR obtains the lowest mean in all five measures: overall MRE is 3.24 pixels, body MRE is 5.33 pixels, face MRE is 1.46 pixels, and left- and right-hand MRE are 5.15 and 5.41 pixels. On CSL-Daily, PEAR has the lowest overall, body, face, and right-hand means (4.65, 5.70, 2.21, and 7.65 pixels), whereas AiOS has the lowest left-hand mean (18.50 pixels, compared with 18.84 for PEAR). The left-hand standard deviations are large: 16.42 pixels for AiOS and 23.61 for PEAR. These descriptive means do not establish a statistically significant difference between the two methods.

On LSCh (OF-HMR), PEAR again obtains the lowest means across all five measures, with overall MRE of 8.19 pixels, body MRE of 10.29 pixels, face MRE of 5.09 pixels, and left- and right-hand MRE of 11.36 and 11.97 pixels. Anatomical differences remain visible within a method: Multi-HMR has left-hand MRE of 32.04 pixels and right-hand MRE of 19.64 pixels, while its face MRE is 7.84 pixels. Overall error therefore does not fully describe agreement in each hand. Because image resolutions and recording conditions differ, pixel errors are compared within each dataset. The results describe automatic 2D reference agreement on the evaluated sequences.

Table 2: PHOENIX-2014-T: source-image reprojection error across ten HMR methods. 42 sequences; 4,568 source frames; 210 × 260 pixels; 25 frames/s. Values are equal-sequence means ± sample standard deviations in pixels. Lower is better.

Method	MRE	MRE body	MRE face	MRE left hand	MRE right hand
AiOS	4.98 ± 1.01	7.66 ± 1.55	2.66 ± 0.61	6.72 ± 1.93	8.36 ± 2.61
HybrIK-X	7.81 ± 1.66	9.54 ± 2.83	4.59 ± 1.09	10.10 ± 3.37	13.29 ± 3.61
Multi-HMR	6.04 ± 1.00	10.27 ± 3.18	2.51 ± 0.68	10.48 ± 3.60	9.97 ± 1.60
OSX	7.69 ± 1.15	10.48 ± 1.58	4.99 ± 1.66	10.45 ± 2.61	11.19 ± 1.81
PEAR	$\mathbf{3.24 \pm 0.53}$	$\mathbf{5.33 \pm 1.06}$	$\mathbf{1.46 \pm 0.16}$	$\mathbf{5.15 \pm 1.54}$	$\mathbf{5.41 \pm 1.19}$
PyMAF-X	7.83 ± 1.57	8.67 ± 1.73	3.75 ± 0.64	12.66 ± 4.51	13.40 ± 3.51
SMPLer-X	7.41 ± 1.17	9.58 ± 1.75	3.46 ± 1.02	11.30 ± 2.17	12.91 ± 2.53
SMPLest-X	7.16 ± 1.23	9.46 ± 1.94	4.33 ± 1.27	9.32 ± 2.22	11.45 ± 2.15
Video-to-SMPL-X	7.58 ± 1.16	8.34 ± 1.73	3.18 ± 0.64	12.11 ± 2.82	14.19 ± 2.79
WHAC	7.42 ± 1.31	11.71 ± 2.19	4.22 ± 1.01	9.82 ± 2.60	11.85 ± 2.65

Evaluation: Original-image RTMW-x landmarks versus corresponding projected mesh landmarks in the rendered reconstruction. All methods use the same sequences and eligible reference observations. Overall MRE covers 99 landmarks: six main-body landmarks, 51 internal facial landmarks, and 21 landmarks per hand.

Regions: Body comprises shoulders, elbows, and wrists only; facial and articulated finger landmarks are excluded. Facial contours and coarse head landmarks are excluded throughout. **Bold** identifies the lowest mean in each column.

Table 3: CSL-Daily: source-image reprojection error across ten HMR methods. 29 sequences; 1,596 source frames; 512 × 512 pixels; 30 frames/s. Values are equal-sequence means ± sample standard deviations in pixels. Lower is better.

Method	MRE	MRE body	MRE face	MRE left hand	MRE right hand
AiOS	6.30 ± 1.09	6.99 ± 1.05	3.38 ± 0.92	$\mathbf{18.50 \pm 16.42}$	10.61 ± 2.32
HybrIK-X	9.83 ± 1.96	10.40 ± 2.67	5.26 ± 1.61	32.78 ± 33.25	16.35 ± 3.68
Multi-HMR	8.66 ± 1.84	14.64 ± 6.19	4.16 ± 1.29	32.40 ± 16.40	12.43 ± 2.35
OSX	10.58 ± 1.77	11.54 ± 1.64	5.62 ± 1.39	26.90 ± 18.69	18.50 ± 3.71
PEAR	$\mathbf{4.65 \pm 0.80}$	$\mathbf{5.70 \pm 1.54}$	$\mathbf{2.21 \pm 0.22}$	18.84 ± 23.61	$\mathbf{7.65 \pm 1.95}$
PyMAF-X	8.64 ± 1.90	8.79 ± 1.47	5.18 ± 1.59	25.46 ± 17.60	12.67 ± 3.91
SMPLer-X	7.49 ± 1.16	9.05 ± 1.52	3.54 ± 1.09	19.58 ± 17.24	13.93 ± 2.30
SMPLest-X	7.10 ± 1.16	10.09 ± 1.91	3.92 ± 1.24	19.09 ± 13.61	11.31 ± 2.34
Video-to-SMPL-X	11.69 ± 2.17	13.79 ± 2.60	4.03 ± 1.07	37.97 ± 34.69	24.03 ± 5.26
WHAC	10.80 ± 1.99	13.27 ± 3.08	5.73 ± 2.02	24.16 ± 17.69	18.98 ± 3.26

Evaluation: Original-image RTMW-x landmarks versus corresponding projected mesh landmarks in the rendered reconstruction. All methods use the same sequences and eligible reference observations. Overall MRE covers 99 landmarks: six main-body landmarks, 51 internal facial landmarks, and 21 landmarks per hand.

Regions: Body comprises shoulders, elbows, and wrists only; facial and articulated finger landmarks are excluded. Facial contours and coarse head landmarks are excluded throughout. Left-hand MRE uses the 26 sequences with eligible left-hand observations; the remaining measures use all 29 sequences. **Bold** identifies the lowest mean in each column.

Table 4: LSCh (OF-HMR): source-image reprojection error across ten HMR methods. 8 sequences; 838 source frames; 1440 × 1080 pixels; 30 frames/s. Values are equal-sequence means ± sample standard deviations in pixels. Lower is better.

Method	MRE	MRE body	MRE face	MRE left hand	MRE right hand
AiOS	11.29 ± 2.34	12.81 ± 3.79	5.24 ± 0.29	19.31 ± 9.34	17.54 ± 3.10
HybrIK-X	17.12 ± 2.03	16.42 ± 3.03	12.85 ± 2.73	20.18 ± 5.97	24.64 ± 5.33
Multi-HMR	16.16 ± 5.93	19.14 ± 7.01	7.84 ± 2.46	32.04 ± 20.48	19.64 ± 2.61
OSX	16.63 ± 2.27	19.94 ± 2.39	7.69 ± 2.06	25.96 ± 8.78	28.07 ± 5.44
PEAR	$\mathbf{8.19 \pm 0.55}$	$\mathbf{10.29 \pm 3.21}$	$\mathbf{5.09 \pm 0.71}$	$\mathbf{11.36 \pm 2.42}$	$\mathbf{11.97 \pm 1.18}$
PyMAF-X	16.04 ± 3.15	16.60 ± 1.69	11.33 ± 2.39	22.47 ± 8.92	20.90 ± 4.09
SMPLer-X	13.28 ± 1.79	15.14 ± 2.35	6.04 ± 0.76	21.00 ± 8.18	22.60 ± 2.47
SMPLest-X	11.37 ± 1.02	18.11 ± 2.62	5.49 ± 0.71	16.88 ± 4.28	18.23 ± 2.82
Video-to-SMPL-X	24.03 ± 10.75	24.47 ± 5.95	9.10 ± 4.53	48.57 ± 34.55	35.63 ± 9.00
WHAC	16.81 ± 4.48	25.09 ± 3.05	8.44 ± 2.06	26.05 ± 14.17	25.51 ± 6.76

Evaluation: Original-image RTMW-x landmarks versus corresponding projected mesh landmarks in the rendered reconstruction. All methods use the same sequences and eligible reference observations. Overall MRE covers 99 landmarks: six main-body landmarks, 51 internal facial landmarks, and 21 landmarks per hand.

Regions: Body comprises shoulders, elbows, and wrists only; facial and articulated finger landmarks are excluded. Facial contours and coarse head landmarks are excluded throughout. **Bold** identifies the lowest mean in each column.

References

[1] W. Sandler, D. Lillo-Martin, Sign Language and Linguistic Universals, Cambridge University Press, 2006. doi:10.1017/CBO9781139163910.

[2] M. Toshpulatov, W. Lee, J. Jun, S. Lee, Deep learning pathways for automatic sign language processing, Pattern Recognition 164 (2025) 111475. doi:10.1016/j.patcog.2025.111475.

[3] S. Tan, N. Khan, Z. An, Y. Ando, R. Kawakami, K. Nakadai, A review of deep learning-based approaches to sign language processing, Advanced Robotics 38 (23) (2024) 1649–1667. doi:10.1080/01691864.2024.2442721.

[4] Z. Yang, S. Zhang, X. Wang, H. Jin, L. Sun, Lightweight whole-body mesh recovery with joints and depth aware hand detail optimization, Journal of Visual Communication and Image Representation 116 (2026) 104729. doi:10.1016/j.jvcir.2026.104729.

[5] Y. Tian, H. Zhang, Y. Liu, L. Wang, Recovering 3d human mesh from monocular images: A survey, IEEE Transactions on Pattern Analysis and Machine Intelligence 45 (12) (2023) 15406–15425. doi:10.1109/TPAMI.2023.3298850.

[6] R. Jayaswal, M. A. Ansari, A. Mewada, P. Pareek, S. Ahmad, An in-depth exploration of structural pose estimation strategies and datasets, Discover Computing 28 (2025) 222. doi:10.1007/s10791-025-09726-8.

[7] K. Yin, A. Moryossef, J. Hochgesang, Y. Goldberg, M. Alikhani, Including signed languages in natural language processing, in: C. Zong, F. Xia, W. Li, R. Navigli (Eds.), Proceedings of the 59th Annual Meeting of the Association for Computational Linguistics and the 11th International Joint Conference on Natural Language Processing (Volume 1: Long Papers), Association for Computational Linguistics, Online, 2021, pp. 7347–7360. doi:10.18653/v1/2021.acl-long.570.
URL https://aclanthology.org/2021.acl-long.570/

[8] Y. Liu, C. Qiu, Z. Zhang, Deep learning for 3d human pose estimation and mesh recovery: A survey, Neurocomputing 596 (2024) 128049. doi:10.1016/j.neucom.2024.128049.

[9] N. C. Camgöz, S. Hadfield, O. Koller, H. Ney, R. Bowden, Neural sign language translation, in: Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition, 2018.
URL https://www-i6.informatik.rwth-aachen.de/~koller/RWTH-PHOENIX-2014-T/

[10] H. Zhou, W. Zhou, W. Qi, J. Pu, H. Li, Improving sign language translation with monolingual data by sign back-translation, in: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2021, pp. 1316–1325.
URL https://openaccess.thecvf.com/content/CVPR2021/html/Zhou_Improving_Sign_Language_Translation_With_Monolingual_Data_by_Sign_Back-Translation_CVPR_2021_paper.html

[11] T. Jiang, X. Xie, Y. Li, RTMW: Real-time multi-person 2D and 3D whole-body pose estimation, arXiv preprint arXiv:2407.08634 (2024). doi:10.48550/arXiv.2407.08634.
URL https://arxiv.org/abs/2407.08634

[12] K. B. Diallo, M. A. Akhloufi, Deep human pose estimation: A conceptual review of paradigms, progress, and frontiers, Computers 15 (6) (2026) 366. doi:10.3390/computers15060366.
URL https://www.mdpi.com/2073-431X/15/6/366

[13] G. Pavlakos, V. Choutas, N. Ghorbani, T. Bolkart, A. A. A. Osman, D. Tzionas, M. J. Black, Expressive body capture: 3D hands, face, and body from a single image, in: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2019, pp. 10975–10985. doi:10.1109/CVPR.2019.01123.
URL https://arxiv.org/abs/1904.05866

[14] H. Cuevas Velasquez, A. Yiannakidis, S. Shin, G. Becherini, M. Höschle, J. Tesch, T. Obersat, T. Alexiadis, E. Halilaj, M. J. Black, MAMMA: Markerless Accurate Multi-person Motion Acquisition, in: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2026, pp. 7175–7186.
URL https://arxiv.org/abs/2506.13040

[15] MotrixLab, Mka: Markerless kinematic analysis (mka), https://github.com/MotrixLab/MKA.git, software, commit 7ebafd38c75c128fe208512acb7226c1b6fb136b. accessed 2026-09-04 (s.f.).

[16] Y. Yoshiyasu, L. Sun, R. Sagawa, Meshmamba: State space models for articulated 3d mesh generation and reconstruction, in: Proceedings of the IEEE/CVF International Conference on Computer Vision, 2025, pp. 6563–6574.

[17] X. Yang, D. Kukreja, D. Pinkus, T. Fan, J. Park, S. Shin, J. Cao, J.-W. Liu, N. Ugrinovic, A. Sagar, J. Malik, M. Feiszli, P. Dollar, K. Kitani, Sam 3d body: Robust full-body human mesh recovery, in: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2026, pp. 7209–7219.

[18] H. E. Pang, Z. Cai, L. Yang, Q. Tao, Z. Wu, T. Zhang, Z. Liu, Towards robust and expressive whole-body human pose and shape estimation, in: Advances in Neural Information Processing Systems, 2023.

[19] S. Heo, Y. Oh, Y. Cha, Smflix: A unified body–head model for real-time expressive human mesh

recovery, Mathematics 14 (18) (2026) 3287. doi:10.3390/math14183287.

[20] H. Zhang, Y. Tian, Y. Zhang, M. Li, L. An, Z. Sun, Y. Liu, Pymaf-x: Towards well-aligned full-body model regression from monocular images, IEEE Transactions on Pattern Analysis and Machine Intelligence 45 (10) (2023) 12287–12303. doi:10.1109/TPAMI.2023.3271691.

[21] J. Lin, A. Zeng, H. Wang, L. Zhang, Y. Li, One-stage 3d whole-body mesh recovery with component aware transformer, in: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2023, pp. 21159–21168. doi:10.1109/CVPR52729.2023.02027.

[22] Z. Cai, W. Yin, A. Zeng, C. Wei, Q. Sun, Y. Wang, H. E. Pang, H. Mei, M. Zhang, L. Zhang, C. C. Loy, L. Yang, Z. Liu, SMPLer-X: Scaling up expressive human pose and shape estimation, in: Advances in Neural Information Processing Systems, Vol. 36, 2023, pp. 11454–11468. doi:10.52202/075280-0506.

[23] J. Li, S. Bian, C. Xu, Z. Chen, L. Yang, C. Lu, Hybrik-x: Hybrid analytical-neural inverse kinematics for whole-body mesh recovery, IEEE Transactions on Pattern Analysis and Machine Intelligence 47 (4) (2025) 2754–2769. doi:10.1109/TPAMI.2025.3528979.

[24] W. Yin, Z. Cai, R. Wang, A. Zeng, C. Wei, Q. Sun, H. Mei, Y. Wang, H. E. Pang, M. Zhang, L. Zhang, C. C. Loy, A. Yamashita, L. Yang, Z. Liu, SMPLest-X: Ultimate scaling for expressive human pose and shape estimation, IEEE Transactions on Pattern Analysis and Machine Intelligence 48 (2) (2026) 1778–1794. doi:10.1109/TPAMI.2025.3618174.

[25] J. Wu, Y. Liu, L. Lin, Y. Zhu, L. Zhu, J. Li, Y. Li, Pear: Pixel-aligned expressive human mesh recovery, in: Proceedings of the Special Interest Group on Computer Graphics and Interactive Techniques Conference Conference Papers, SIGGRAPH '26 Conference Papers, Association for Computing Machinery, 2026, pp. 46:1–46:12. doi:10.1145/3799902.3811096.

[26] Q. Sun, Y. Wang, A. Zeng, W. Yin, C. Wei, W. Wang, H. Mei, C.-S. Leung, Z. Liu, L. Yang, Z. Cai, Aios: All-in-one-stage expressive human pose and shape estimation, in: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2024, pp. 1834–1843. doi:10.1109/CVPR52733.2024.00180.

[27] F. Baradel, M. Armando, S. Galaaoui, R. Brégier, P. Weinzaepfel, G. Rogez, T. Lucas, Multi-hmr: Multi-person whole-body human mesh recovery in a single shot, in: European Conference on Computer Vision, 2024. doi:10.1007/978-3-031-73337-6_12.

[28] W. Yin, Z. Cai, R. Wang, F. Wang, C. Wei, H. Mei, W. Xiao, Z. Yang, Q. Sun, A. Yamashita, Z. Liu, L. Yang, Whac: World-grounded humans and cameras, in: European Conference on Computer Vision, 2024, pp. 20–37. doi:10.1007/978-3-031-72754-2_2.

[29] articulab, Video-to-smpl-x: vid2smplx, https://github.com/articulab/vid2smplx.git, software, commit dbb8f377c12381136fe0820939de1c0bc8494e6b. accessed 2026-09-04 (s.f.).